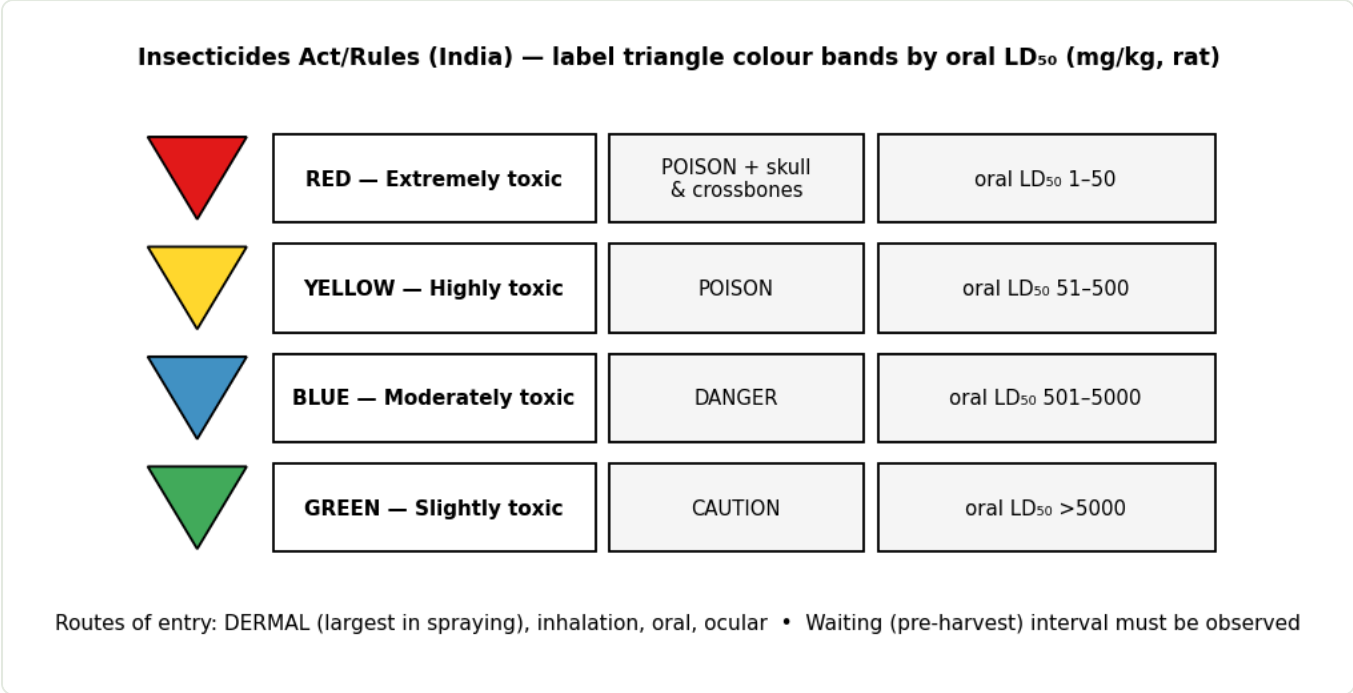
UNIT 5 · TOPIC 7

Chemical Hazards and Control Measures

Syllabus topic: Chemical hazards and control measures

Toxicity classification (Insecticides Act 1968 / Rules 1971)



Insecticides Act/Rules — label triangle colour bands by oral LD_{50} (mg/kg, rat)

Label triangle	Class	Oral LD_{50} (mg/kg body weight)	Signal
RED	Extremely toxic	1–50	Skull & crossbones + POISON
YELLOW	Highly toxic	51–500	POISON
BLUE	Moderately toxic	501–5000	DANGER
GREEN	Slightly toxic	>5000	CAUTION

- **LD_{50}** = the dose that kills 50% of test animals. **LOWER LD_{50} = MORE toxic.**
- WHO classes Ia / Ib / II / III run in parallel.
- India has banned or restricted several molecules (e.g., monocrotophos restrictions, phased bans) — check the current CIB&RC list.

Routes of entry and health effects

Route	Notes
DERMAL	The largest route during spraying — hands and back
Inhalation	Fine mists, fumigants
Oral	Accidental, or eating/smoking while spraying
Ocular	Splashes

- **Acute:** cholinesterase inhibition from organophosphates/carbamates — headache, salivation, miosis, vomiting, convulsions. **Atropine** is the antidote; treat as an emergency.
- **Chronic** effects follow repeated exposure.

Control measures — engineering and practice

1. Choose the least-toxic effective molecule and the correct dose (calibration — Unit 3, Topic 1).
2. Use closed mixing/transfer; mix only what is needed; avoid dusts and mist-blowers where drift is likely.
3. Use low-drift nozzles at the proper pressure (Unit 3, Topic 3b).
4. Spray with the wind at your back, moving out of the treated zone.
5. Never blow a blocked nozzle by mouth — use a pin/brush.
6. Triple-rinse and puncture empty containers; never reuse them.
7. Store chemicals locked, away from food, feed and children.
8. Observe the WAITING (pre-harvest) INTERVAL and the re-entry period.
9. Bathe after spraying and launder spray clothes separately.
10. Keep clean water, soap and first aid on site.

Drone/aerial spraying adds buffer-zone and PPE rules (Unit 2, Topic 12).

Practice MCQs

Q93. LD_{50} means the dose that:

(a) Cures 50% (b) Kills 50% of test animals
(c) Is 50 mg (d) Lasts 50 days

Q94. A LOWER LD_{50} value indicates the chemical is:

(a) Safer (b) More toxic
(c) Cheaper (d) Less volatile

Q95. The RED triangle label denotes oral LD_{50} of:

(a) 1–50 mg/kg (extremely toxic) (b) 51–500
(c) 501–5000 (d) >5000

Q96. The YELLOW triangle corresponds to:

(a) Extremely toxic (b) Highly toxic, LD_{50} 51–500
(c) Moderately toxic (d) Slightly toxic

Q97. The BLUE triangle signal word is:

(a) POISON (b) DANGER
(c) CAUTION (d) SAFE

Q98. The GREEN triangle indicates oral LD_{50}:

(a) <50 (b) 51–500
(c) 501–5000 (d) >5000 mg/kg

Q99. Pesticide labelling classes in India are prescribed under the:

(a) Insecticides Act/Rules (b) Factories Act
(c) Motor Vehicles Act (d) BIS Act

Q100. The MAJOR route of pesticide entry during spraying is:

(a) Oral (b) Dermal (skin)
(c) Ocular (d) Rectal

Q101. Antidote for organophosphate poisoning is:

(a) Atropine
(b) Glucose
(c) Vitamin C
(d) Antihistamine

Q102. Blocked spray nozzles should be cleared:

(a) By blowing with the mouth
(b) With a pin/brush or water
(c) With petrol
(d) With the tongue

Q104. The waiting (pre-harvest) interval is observed to:

(a) Save chemical
(b) Ensure residues fall below safe limits before harvest
(c) Improve yield
(d) Reduce cost

Answers

Q93 — (b) Kills 50% of test animals
Q94 — (b) More toxic — *LD_{50} is an inverse measure of toxicity.*
Q95 — (a) 1–50 mg/kg (extremely toxic) — *The four label triangles are a guaranteed-recall item.*
Q96 — (b) Highly toxic, LD_{50} 51–500
Q97 — (b) DANGER
Q98 — (d) >5000 mg/kg
Q99 — (a) Insecticides Act/Rules
Q100 — (b) Dermal (skin) — *Skin absorption dominates in knapsack spraying, so gloves/overalls matter more than masks in many cases.*
Q101 — (a) Atropine
Q102 — (b) With a pin/brush or water
Q104 — (b) Ensure residues fall below safe limits before harvest

Sources: Liljedahl et al. — Tractors and their Power Units, Ch. 9, pp. 231–235; ASAE S383.1 & S310.3; BIS safety standards IS 9020 / IS 9129 / IS 12239; Insecticides Act 1968 & Rules 1971 (label classes); WHO hazard classification; ICAR-CIAE / AICRP farm-accident surveys; NPTEL lectures.